

LaRT v2.00 User Manual

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1. Introduction

LaRT (**L**ayman-**a**lpha **R**adiative **T**ransfer) is a Monte Carlo radiative transfer code for computing the propagation of resonance-line photons through astrophysical neutral-gas media. The default transition is Lyman- α ($\text{Ly}\alpha$, $\text{H I } 1216 \text{ \AA}$), but many other UV/optical/IR lines are supported (see §5.3). Dust scattering and absorption, full Stokes polarization tracking, Hubble-flow and other bulk velocity fields, and spatially varying density, temperature, and velocity from external data files are all supported.

Starting from v2.00, the Cartesian grid is supplemented by:

- **AMR mode** — octree adaptive mesh refinement, allowing spatially varying resolution (§7.).
- **Clumpy medium mode** — a population of compact, optically thick spherical clumps embedded in a transparent inter-clump medium (§8.).

This manual covers v2.00 and v2.10. The only user-visible change in v2.10 is the introduction of `par%grid_type` as a unified selector for the three grid modes; the old `par%use_amr_grid` and `par%use_clump_medium` flags remain valid for backward compatibility.

2. Installation and Build

2.1 Prerequisites

- Fortran 2003+ compiler: Intel `ifort/ifx` or GNU `gfortran` ≥ 7 .
- CFITSIO library.
- HDF5 library (Fortran bindings), version ≥ 1.14 , built with the same compiler used for LaRT. Required for the default build (`HDF5=1`); skip with `HDF5=0` for a FITS-only build.
- MPI library (MPICH or OpenMPI).

2.2 Compilation

```
cd LaRT_v2.00          # or LaRT_v2.10
make                   # produces LaRT.x (HDF5=1 is the default)
make HDF5=0           # FITS-only build (no HDF5 link dependency)
make HDF5_PREFIX=/path # override HDF5 install location
make DEBUG=1          # full debug flags + runtime checks
make F90=mpif90        # override compiler
make clean             # remove .o and .mod files
```

```
make cleanall          # also remove executables
make convert_ramses    # auxiliary RAMSES converter
```

Every make invocation recompiles all objects (no incremental builds). `par%HDF5_PREFIX` defaults to `/data/opt/hdf5_intel` on this machine; set it on the command line or as an environment variable when building elsewhere.

2.3 Running

```
mpirun -np 8 LaRT.x input.in
```

Output is written to `<base>.h5` by default (or `<base>.fits.gz` when `par%file_format = 'fits'` is set), where `<base>` is the input file base name (without `.in`) unless `par%out_file` is set explicitly. See §11. for the two supported file formats and how to choose between them.

3. Input File Format

Input files use Fortran *namelist* syntax. All parameters are prefixed with `par%` and collected in a single `¶meters` block.

```
1 &parameters
2   par%no_photons      = 1e6
3   par%temperature     = 1.0e4      ! gas temperature [K]
4   par%taumax          = 1.0e4
5   par%nx = 101, par%ny = 101, par%nz = 101
6   par%rmax            = 1.0
7   par%source_geometry = 'point'
8   par%spectral_type   = 'voigt'
9   par%nxfreq          = 121
10 /
```

- String values must be enclosed in single quotes.
- Multiple parameters may appear on one line, separated by commas.
- Lines beginning with `!` are comments.
- Unspecified parameters retain compiled defaults (listed in §14.).

4. Grid Configuration

4.1 Cartesian Grid

The simulation domain is a rectangular box centered at the origin. By default $x \in [-x_{\max}, +x_{\max}]$ (and similarly for y and z). The cell size is $\Delta x = 2x_{\max}/n_x$.

Parameter	Default	Description
par%nx, par%ny, par%nz	101	Number of cells per axis
par%xmax, par%ymax, par%zmax	1.0	Half-box size per axis (code units)
par%rmax	-999	Spherical boundary: density = 0 for $r > r_{\max}$
par%rmin	-999	Inner cavity: density = 0 for $r < r_{\min}$

If par%input_field or par%dens_file is given, (nx, ny, nz) and (xmax, ymax, zmax) are read from the FITS header automatically.

4.2 Grid Geometry

par%geometry selects the spatial symmetry used for accumulating and saving the mean-intensity array J :

Value	J output geometry
'sphere' (default)	1-D radial profile $J(r)$
'cylinder'	2-D cylindrical $J(\rho, z)$
'rectangle'	3-D Cartesian cube $J(x, y, z)$
'plane_atmosphere'	1-D slab $J(z)$
'spherical_atmosphere'	1-D spherical shell $J(r)$

4.3 Symmetry and Periodic Boundaries

Parameter	Default	Effect
par%xyz_symmetry	.false.	Simulate only the +x, +y, +z octant (8× RAM saving)
par%xy_symmetry	.false.	Simulate only the +x, +y quadrant (4× RAM saving)
par%xy_periodic	.false.	Periodic in x, y: photons exiting one face re-enter the opposite

Symmetry options are disabled automatically when peel-off is active.

4.4 Distance Units

Parameter	Default	Description
par%distance_unit	' '	Physical unit: 'kpc', 'pc', 'au', or ' ' (dimensionless)
par%distance2cm	1.0	Alternatively: 1 code unit in cm

In dimensionless mode (par%distance_unit = ' '), the gas density is determined entirely by par%taumax and the grid geometry; no physical unit is needed.

4.5 Grid Type Selection

Parameter	Default	Description
par%grid_type ⁱ	'car'	v2.10+ : 'car' (Cartesian), 'amr', or 'clump'
par%use_amr_grid	.false.	v2.00 : enable AMR (deprecated in v2.10)
par%use_clump_medium	.false.	v2.00 : enable clump mode (deprecated in v2.10)

5. Source Configuration

5.1 Source Geometry

par%source_geometry controls the spatial distribution of photon emission:

Value	Description
'point'	Point source at (par%xs_point, par%ys_point, par%zs_point); default (0,0,0)
'uniform'	Uniform volume emissivity throughout the grid (or inside par%source_rmax)
'uniform_sphere'	Uniform within a sphere of radius par%source_rmax
'uniform_xy'	Uniform in the $z = 0$ plane
'gaussian'	Gaussian: $\exp(-r^2/2\sigma_r^2)\exp(-z^2/2\sigma_z^2)$; widths par%source_rscale, par%source_zscale
'exponential'	Exponential disc: $\exp(-r/r_s)\exp(- z /z_s)$; scales par%source_rscale, par%source_zscale
'exponential_sphere'	Exponential clipped to sphere of radius par%source_rmax
'exponential_cylinder'	Exponential clipped to a cylinder
'sersic' / 'ssh'	Sérsic profile; index par%sersic_m, half-light radius par%Reff
'diffuse_emissivity'	3-D emissivity read from par%emiss_file
'star_file'	Individual point sources from text file par%star_file (columns: x, y, z, L)

Associated parameters:

Parameter	Default	Description
par%xs_point, par%ys_point, par%zs_point	0.0	Point source position (code units)
par%source_rmax	-999	Outer truncation radius for extended sources
par%source_rscale	-999	Radial scale length
par%source_zscale	0.0	Vertical scale height
par%sersic_m	4.0	Sérsic index
par%Reff	1.0	Effective (half-light) radius
par%comoving_source	.true.	Source photons co-moving with the local gas

5.2 Spectral Type

par%spectral_type sets the frequency distribution of emitted photons:

Value	Description
'monochromatic'	Delta function at line center ($x = 0$)
'voigt'	Voigt profile at the local gas temperature
'voigt0'	Voigt profile at the fixed temperature par%temperature0
'gaussian'	Gaussian with width par%gaussian_width_vel [km s ⁻¹]
'continuum'	Flat spectrum over [par%xfreq_min, par%xfreq_max]
'line_prof_file'	Arbitrary profile read from par%line_prof_file

5.3 Line Identification

The atomic transition is selected with par%line_id (default: Ly α).

par%line_id	Transition	Wavelength
(default)	H I Ly α	1215.67 Å
'ly_alpha_HD'	H I + D I Ly α	1215.67 Å (see below)
'CIV_1548'	C IV doublet	1548.19 / 1550.77 Å
'NV_1239'	N V doublet	1238.82 / 1242.80 Å
'OVI_1032'	O VI doublet	1031.91 / 1037.61 Å
'SiIV_1394'	Si IV doublet	1393.76 / 1402.77 Å
'NaI_D'	Na I D doublet	5891.58 / 5897.56 Å
'CaII_HK'	Ca II H & K doublet	3934.78 / 3969.59 Å
'MgII_2796'	Mg II doublet	2796.35 / 2803.53 Å
'AlII_1671'	Al II singlet	1670.79 Å
'SiII_1190' / 'SiII_1193'	Si II doublet (+ fluorescence)	1190.42 / 1193.29 Å
'SiII_1260'	Si II (+ fluorescence)	1260.42 Å
'SiII_1304'	Si II (+ fluorescence)	1304.37 Å
'SiII_1527'	Si II (+ fluorescence)	1526.71 Å
'FeII_2250'	Fe II (+ fluorescence)	2249.88 Å
'FeII_2261'	Fe II (+ fluorescence)	2260.78 Å
'FeII_UV3' / 'FeII_2344'	Fe II UV3 multiplet	2344.21 Å
'FeII_UV2' / 'FeII_2383'	Fe II UV2 multiplet	2382.76 Å
'FeII_UV1' / 'FeII_2600'	Fe II UV1 multiplet	2600.17 Å
'HeI_10833'	He I triplet	10832.06 Å

The numerical line%line_type that controls the dispatch of par%calc_voigt / par%do_resonance (set internally by setup_resonance_line) takes the following values: 1 = singlet (one upward transition); 2 = doublet (two upward

transitions sharing a single thermal width, e.g. C IV, N V, O VI, Na I D, Ca II H&K, Mg II, Si IV); 4 = one resonance plus one or more fluorescent decays (most Si II, Fe II UV3); 5 = multiple upward + multiple downward transitions, the most general case (Si II 1190/1193, Fe II UV1, Fe II UV2); 6 = three upward transitions plus one downward transition (He I 10833); 7 = combined H + D Ly α (ly_alpha_HD, two coexisting two-level resonance scatterers, no fine structure).

5.3.1 Combined Hydrogen + Deuterium Ly α

The line id 'ly_alpha_HD' treats neutral hydrogen and deuterium Ly α as two coexisting two-level resonance scatterers in the same medium. The two transitions sit only 0.33 Å (~ 82 km/s) apart and have nearly identical atomic constants except for the atomic mass and rest wavelength. The deuterium-to-hydrogen number-abundance ratio is set by

Parameter	Default	Description
par%D_to_H_ratio	1.5×10^{-5}	n_D/n_H (cosmic primordial)
par%include_deuterium	.false.	Convenience flag

Setting par%include_deuterium = .true. together with par%line_id = 'ly_alpha' promotes the run automatically to 'ly_alpha_HD', so existing input files can opt in without changing the line id.

Physical assumptions in this version (v1):

- Both H and D are treated as simple two-level resonance scatterers (line_type=7). Fine-structure splitting is *not* included for either species; if par%fine_structure = .true. is set together with 'ly_alpha_HD', the code emits a warning and disables fine structure.
- The phase function is the simple dipole ($E_1 = 1, E_2 = 0, E_3 = 1$), identical for both species.
- Oscillator strengths and damping rates are taken to be equal for H and D ($f_D = f_H, \Gamma_D = \Gamma_H$ to better than 0.1%). Mass and rest wavelength differences are fully accounted for in the Voigt evaluation and species selection.
- Photon xfreq is carried in H-frame Doppler units throughout. D-frame conversions happen on-the-fly in the combined Voigt and scattering routines via constants precomputed at line setup (line%delta_nu_HD_Hz, line%ratio_Dfreq_HD, line%ratio_voigta_HD). No per-cell deuterium arrays are stored.
- During a deuterium scattering event, the perpendicular thermal velocity components and the recoil shift are evaluated using the H atomic constants, which produces a small ($\sim 1/\sqrt{2}$) error in the re-emission shape from D-scattered photons. Because D scatters are rare at cosmic abundance and the dominant D feature (the absorption dip) depends on opacity rather than re-emission shape, this is acceptable for v1. Per-photon species tracking is also not yet implemented.

The Deuterium Ly α rest wavelength used is $\lambda_D = 1215.337$ (NIST vacuum), ~ 0.33 blueward of $\lambda_H = 1215.668$. In LaRT's velocity convention ($v = -c \Delta\lambda/\lambda_H$, so negative v is blueshifted), the D Ly α line center sits at $v_D \approx -82$ km/s. A reference example with five validation runs and a Jupyter notebook is provided under examples/lya_HD/ (see run.sh and inspect_HD.ipynb). One of those runs is configured for direct comparison against Dijkstra, Haiman & Spaans (2006), ApJ 649, 14, Figure 3 (D/H = 3×10^{-5} , $\tau \sim 10^6$).

5.4 Frequency and Wavelength Grid

Internally, photon frequencies are tracked in the dimensionless Doppler variable:

$$x = \frac{\nu - \nu_0}{\Delta\nu_D}, \quad \Delta\nu_D = \frac{\nu_0}{c} \sqrt{\frac{2kT}{m}}, \quad (1)$$

where ν_0 is the line-center frequency, T is the local gas temperature, and m is the atomic/ionic mass. Output spectra are binned onto a uniform grid in x by default. A wavelength or velocity grid can be requested instead:

Parameter	Default	Description
par%nxfreq	121	Number of spectral bins
par%xfreq_min, par%xfreq_max	auto	Limits of x -grid
par%xfreq0	0.0	Frequency offset applied to all emitted photons
par%nwavelength	0	Number of wavelength bins (overrides x -grid)
par%wavelength_min, par%wavelength_max	—	[Å]
par%nvelocity	0	Number of velocity bins (overrides x -grid)
par%velocity_min, par%velocity_max	—	[km s ⁻¹]

The range is set automatically from the mean medium temperature when no limits are provided.

6. Medium Properties

6.1 Optical Depth Normalization

The overall opacity is scaled by one of the following (specify at most one):

Parameter	Description
par%taumax	Line-center τ_0 from the grid center to the $+z$ boundary along the z -axis
par%tauhomo	Same, assuming a uniform density field (scales an inhomogeneous field proportionally)
par%N_HI_max	H I column density from center to $+z$ boundary [cm ⁻²]
par%N_HI_homo	Same, assuming uniform density

For AMR grids, par%taumax is the pole-to-face optical depth (box center $\rightarrow +z$ face along z), consistent with the Cartesian convention for a sphere of radius $R = L/2$.

6.2 Temperature

Parameter	Default	Description
par%temperature	10 ⁴ K	Uniform gas temperature
par%temperature0	= T	Temperature for spectral_type='voigt0' source
par%bturb	0.0	Turbulent velocity dispersion added in quadrature to thermal width [km s ⁻¹]
par%temp_file	''	3-D temperature cube [K]; FITS or HDF5

6.3 Density

Analytic profiles. Without external files, the density is set analytically via the opacity scale parameters (par%taumax etc.) combined with an optional radial profile:

Profile type	Key parameter	Functional form
Uniform (sphere or box)	par%rmax	Constant n_{H} inside $r < r_{\text{max}}$
Spherical shell	par%rmin, par%rmax	Constant n_{H} between r_{min} and r_{max}
Exponential	par%density_rscale	$n \propto \exp(-r/r_s)$

External gridded data (FITS or HDF5).

The reader (read_grid_data.f90) detects the format from the filename extension (.fits[.gz], .h5, or .hdf5) and uses the same code path for both. HDF5 inputs expect the array at /<section>/data (any section name); shape information is read from the dataset itself, so FITS-style NAXIS keywords are not required.

Parameter	Description
par%dens_file	3-D H I number density cube [cm^{-3}]
par%temp_file	3-D temperature cube [K]
par%velo_file	4-D velocity cube (v_x, v_y, v_z) [km s^{-1}]
par%emiss_file	3-D emissivity cube [$\text{photons s}^{-1} \text{cm}^{-3}$]
par%input_field	Base name: reads <base>.dens.fits.gz, .temp., .velo. (legacy convention; FITS-only)

If par%taumax (or par%N_HI_max) is specified alongside a density file, the density is rescaled so that the target optical depth is achieved.

6.4 Velocity Fields

par%velocity_type selects the bulk velocity model. The default (empty string) is a static medium.

Value	Key parameters	Velocity law
'hubble'	par%Vexp	$\mathbf{v}(r) = V_{\text{exp}}(r/r_{\text{max}})\hat{r}$; $V_{\text{exp}} > 0$: outflow
'constant_radial'	par%Vexp	$\mathbf{v}(r) = V_{\text{exp}}\hat{r}$ (constant magnitude)
'parallel_velocity'	par%Vx, par%Vy, par%Vz	Uniform $\mathbf{v} = (V_x, V_y, V_z)$ [km s^{-1}]
'ssh'	par%Vpeak, par%rpeak, par%DeltaV	Bilinear profile (Song, Seon & Hwang 2020)
'rotating_solid_body'	par%Vrot, par%rinner	Solid-body rotation
'rotating_galaxy_halo'	par%Vrot, par%q	Galaxy halo rotation curve

The 'ssh' profile rises linearly to V_{peak} at r_{peak} , then shifts by ΔV toward r_{max} :

$$v(r) = \begin{cases} V_{\text{peak}} r / r_{\text{peak}} & r < r_{\text{peak}} \\ \left[V_{\text{peak}} + \Delta V \frac{r - r_{\text{peak}}}{r_{\text{max}} - r_{\text{peak}}} \right] \hat{r} & r \geq r_{\text{peak}}. \end{cases} \quad (2)$$

A velocity FITS file (par%velo_file) may be combined with a systematic par%velocity_type.

Other velocity-related parameters:

Parameter	Default	Description
par%Omega	28.0	Shear rate [(km s ⁻¹) kpc ⁻¹] for TIGRESS data
par%recoil	.false.	Include photon-recoil frequency shift in scattering
par%core_skip	.false.	Enable core-skipping acceleration algorithm

7. AMR Grid Mode

7.1 Enabling AMR Mode

v2.00:

```

1 par%use_amr_grid = .true.
2 par%amr_type     = 'generic'    ! always specify explicitly
3 par%amr_file     = 'grid.dat'
```

v2.10 (preferred):

```

1 par%grid_type = 'amr'
2 par%amr_type  = 'generic'
3 par%amr_file  = 'grid.dat'
```

Parameter	Default	Description
par%amr_type	'ramses'	Format: 'generic' (text) or 'ramses' (binary snapshot)
par%amr_file	''	Path to AMR data file or RAMSES snapshot directory
par%amr_snapnum	1	Snapshot number for RAMSES output

Note: the compiled default for par%amr_type is 'ramses'. Always specify par%amr_type = 'generic' explicitly when using the text format.

7.2 Generic AMR Data Format

Plain text: a header followed by one line per leaf cell.

```

1 # comment lines start with #
2 BOXLEN  100.0           ! box side length in code units
3 NLEAF   8192            ! number of leaf cells
```

```

4 | ! columns: x   y   z   level  nH[cm^-3]  T[K]   vx[km/s]  vy[km/s]  vz[km/s]
5 | 50.5   50.5   50.5   1   1.0e-3   1.0e4   0.0   0.0   0.0
6 | ...

```

Leaf-cell coordinates are given with the box origin at the corner, $[0, \text{BOXLEN}]$. The reader internally re-centers to $[-L/2, +L/2]$; *do not* pre-shift the data.

7.3 Tau Normalization Convention

For AMR, `par%taumax` is the line-center optical depth from the box center to the $+z$ face along the z -axis. This matches the Cartesian sphere convention where `par%taumax` is the center-to-surface optical depth, provided $r_{\text{max}} = L/2$.

7.4 Source Geometries in AMR Mode

In v2.00, AMR supports 'point' and 'uniform'. In v2.10, most Cartesian geometries (Gaussian, exponential, Sérsic, `star_file`, etc.) are also available; refer to §5.1 and the v2.10 release notes.

8. Clumpy Medium Mode

8.1 Overview

Clumpy medium mode places N spherical clumps of radius r_c inside a spherical outer domain of radius R . The inter-clump medium is vacuum. All opacity resides inside the clumps. By default, clump positions are drawn by Random Sequential Adsorption (RSA) so that clumps never overlap. Optionally, overlapping clumps can be generated or loaded from an external FITS file; LaRT detects overlaps automatically after placement and engages an event-based multi-component raytrace in that case (see §8.6).

8.2 Enabling Clump Mode

v2.00:

```

1 | par%use_clump_medium = .true.

```

v2.10 (preferred):

```

1 | par%grid_type = 'clump'

```

8.3 Clump Parameters

Parameter	Default	Description
par%rmax	1.0	Outer sphere radius (code units)
par%rmin	-999	Inner cavity radius (the default sentinel is treated as 0). Clumps are placed only in the shell $r_{\min} \leq r \leq r_{\max}$
par%clump_radius	—	Individual clump radius (same units)
par%clump_tau0	—	Line-center τ_0 from clump center to its surface
par%clump_f_cov	—	Covering factor $C_f = N r_c^2 / R^2$
par%clump_f_vol	—	Volume filling factor $f_V = N (r_c / R)^3$
par%clump_N_clumps	—	Total number of clumps N
par%temperature	10^4 K	Clump gas temperature
par%clump_sigma_v	0.0	Gaussian σ of random bulk clump velocities [km s^{-1}]
par%save_clump_info	.false.	Write clump positions and velocities to a FITS binary table
par%clump_input_file	''	If non-empty, load the clump population from this FITS file (see §8.5) instead of generating it internally
par%clump_fully_inside	.true.	If .true. (default), reject any candidate whose outermost edge would protrude past the outer sphere ($d_{\text{center}} + r_c > R$) or into the inner cavity ($d_{\text{center}} - r_c < r_{\min}$); if .false., restore the legacy behavior in which only the center is required to lie in $[r_{\min}, R]$
par%clump_allow_overlap	.false.	If .true., skip RSA overlap rejection during internal generation: clumps are placed at uniformly random positions. After placement check_has_overlap() is called automatically; if any overlapping pair is found the overlap-aware raytrace is engaged. See §8.6

Specify exactly one of par%clump_f_cov, par%clump_f_vol, or par%clump_N_clumps.

For the per-clump opacity, specify *one* of par%clump_tau0, par%clump_NHI, or par%clump_nH. As a convenience, the system-level radial-sightline targets par%taumax and par%N_HI_max are also accepted: when none of par%clump_tau0/par%clump_NHI/par%clump_nH is set, LaRT back-solves the peak per-clump opacity so the realised radial sightline from the box centre to $r = R$ matches the requested par%taumax (or par%N_HI_max). In the uniform-clump case the back-solve uses the closed-form $\tau_{\max} = \frac{4}{3} f_{\text{cov}} \tau_{c,0}$ (and similarly for $N_{\text{HI,max}}$); in profile mode it uses the radial quadrature derived from par%clump_radius_profile, par%clump_density_profile, and par%clump_number_profile. par%N_gasmax is treated as an alias for par%N_HI_max in clump mode (HI-only line). When a pre-built clump population is loaded via par%clump_input_file, supplying par%taumax or par%N_HI_max multiplies every $\text{cl_rhokap}(i)$ by a single factor α so the realised target is matched (small-angle expected-column formula); the per-clump table is otherwise preserved.

When `par%rmin` is set, `par%clump_f_vol` and `par%clump_f_cov` are interpreted relative to the shell volume $V_{\text{shell}} = \frac{4}{3}\pi(R^3 - r_{\text{min}}^3)$ and the shell sightline factor $R^2 + Rr_{\text{min}} + r_{\text{min}}^2$. With `par%rmin` unset (default -999, treated as 0) these reduce exactly to the original $(R/r_c)^3$ and $(R/r_c)^2$ formulas, preserving prior runs. When `par%clump_fully_inside` is on, the realized filling factors fall slightly below the requested target whenever r_c is a non-trivial fraction of $R - r_{\text{min}}$, since the available placement volume shrinks from $\frac{4}{3}\pi(R^3 - r_{\text{min}}^3)$ to $\frac{4}{3}\pi[(R - r_c)^3 - (r_{\text{min}} + r_c)^3]$. Specify a slightly larger target if exact matching is required. See the *Clumpy Medium Algorithm Description* document, “Fully-Inside Constraint” subsection, for the placement details.

8.4 Spatially-varying clump properties

Three optional radial-profile inputs let the per-clump radius, internal H I density, and spatial number density vary with distance from the box center. Built-in shapes are ‘constant’ (default), ‘powerlaw’, ‘gaussian’, ‘exponential’, and ‘file’ (5-column ASCII table). Setting any flag to a value other than ‘constant’ switches `par%init_clumps` to the profile-aware path.

Parameter	Default	Description
<code>par%clump_radius_profile</code>	‘constant’	shape of $r_c(r)$
<code>par%clump_radius_alpha</code>	0.0	power-law index for $r_c(r)$
<code>par%clump_radius_r0</code>	0.0	power-law scale for $r_c(r)$
<code>par%clump_density_profile</code>	‘constant’	shape of internal $n_H(r)$
<code>par%clump_density_alpha</code>	0.0	power-law index for $n_H(r)$
<code>par%clump_density_r0</code>	0.0	power-law scale for $n_H(r)$
<code>par%clump_number_profile</code>	‘constant’	shape of spatial $n_{cl}(r)$
<code>par%clump_number_alpha</code>	0.0	power-law index for $n_{cl}(r)$
<code>par%clump_number_r0</code>	0.0	power-law scale for $n_{cl}(r)$
<code>par%clump_profile_file</code>	‘’	5-column tabulated profile (used when any axis = ‘file’)

For non-uniform profiles the closed-form $f_{\text{cov}} / f_{\text{vol}} / N$ relation does not hold. Instead the profile-aware quadrature back-solves the spatial normalization so that whichever of `par%clump_f_cov`, `par%clump_f_vol`, or `par%clump_N_clumps` is provided is matched, and the other two are reported on stdout. See the *Clumpy Medium Algorithm Description* document for the integral definitions and the implementation outline.

8.5 Standalone clump generator (make_clumps.x)

The same Fortran routines that build the clump population inside LaRT are also exposed through a small standalone driver, `make_clumps.x`:

```

1 make make_clumps                # builds make_clumps.x
2 mpirun -np N ./make_clumps.x clumps.in  # writes <base>_clumps.fits.gz
```

The driver reads the same namelist as LaRT.x and writes the realized clump population to `<base_name>_clumps.fits.gz` without running the photon transport. Fix `par%iseed` to a non-zero value for reproducibility: under the same seed the table produced by `make_clumps.x` matches the `_clumps.fits.gz` written by a regular LaRT.x run with `par%save_clump_info = .true.` byte-for-byte.

To consume the file, set `par%clump_input_file` on the LaRT input:

```

1 &parameters
2   par%grid_type      = 'clump'
3   par%rmax           = 1.0
4   par%clump_radius   = 0.01
5   par%clump_tau0     = 1.0e3
6   par%temperature    = 1.0e4
7   par%no_photons     = 1e6
8   par%clump_input_file = 'clumps_clumps.fits.gz'
9 /

```

When `par%clump_input_file` is non-empty, `init_clumps` skips the profile-detection and RSA-generation paths entirely and reads the population from the file. `read_clumps_fits` accepts files with any subset of the per-clump columns `R_CLUMP/RHOKAP/TEMP`; missing columns are filled from the corresponding header keyword (`CL_RAD/RHOKAP/TEMP_CL`). For the `RHOKAP` slot the column names `DENSITY` and `DENS` are also accepted as aliases, in priority order `RHOKAP` → `DENSITY` → `DENS`.

8.6 Overlapping Clumps

By default LaRT guarantees non-overlapping clumps through RSA placement and all raytrace paths assume this property. Two situations can produce overlapping populations:

1. A clump population loaded from an external FITS file via `par%clump_input_file` may contain overlapping clumps.
2. Setting `par%clump_allow_overlap = .true.` skips the RSA rejection step so that internally generated clumps are placed at uniformly random positions and will typically overlap whenever the clumps are large relative to the domain.

After the CSR acceleration grid is built, `check_has_overlap()` scans all clump pairs. If any overlapping pair is found it prints

```

1 Overlap check: overlapping clumps detected -- using overlap-aware raytrace.

```

sets an internal flag `has_overlap = .true.`, and routes all subsequent raytrace calls to the overlap-aware variants automatically. No input change is required; the detection and dispatch happen at run time.

Physical model for overlapping clumps

Two clumps sharing a volume with different bulk velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ are treated as two independent gas components coexisting at the same spatial location. The total line opacity at global-frame frequency x is

$$\kappa_{\text{total}}(x) = \sum_i \rho \kappa_i H(x - u_{\text{los},i}, a_i), \quad u_{\text{los},i} = \frac{\mathbf{v}_i \cdot \hat{\mathbf{k}}}{\Delta \nu_{D,i}}. \quad (3)$$

When a scattering event occurs in an overlap region the interacting atom belongs to one clump, chosen by opacity-weighted sampling $P(\text{clump } i) = \kappa_i / \kappa_{\text{total}}$.

Quick-start example

Large clumps with low covering factor guarantee many overlaps while keeping the run fast:

```

1 par%grid_type           = 'clump'
2 par%clump_radius        = 0.3
3 par%clump_f_cov         = 2.0
4 par%clump_allow_overlap = .true.
5 par%no_photons          = 1e5
6 par%spectral_type       = 'voigt'
```

Reference inputs: `clump_overlap_mono.in`, `clump_overlap_sigv.in`, `clump_overlap_peel.in` in `examples/clump_sphere/`.

For a full algorithmic description see the *Clumpy Medium Algorithm Description* document, Section 15 “Overlap-Aware Raytrace”.

8.7 Clump Velocity

A systematic flow can be added on top of the random component (`par%clump_sigma_v`) via `par%velocity_type`. Supported values in clump mode: `'hubble'`, `'constant_radial'`, `'parallel_velocity'`, `'ssh'`, `'rotating_solid_body'`, `'rotating_galaxy_halo'`.

9. Observers and Peel-off

The peel-off (peeling-off) technique shoots a photon toward the observer at every scattering event, allowing spectral images to be constructed without noise. It is activated by setting `par%nxim` and `par%nyim` to non-zero values.

9.1 Observer Placement

Direction vector:

```

1 par%obsx = 0.0, par%obsy = 0.0, par%obsz = 1.0 ! look down +z axis
2 par%distance = 100.0 ! observer-center separation
```

Spherical angles:


```

1 par%alpha = 45.0    ! azimuthal angle [deg]
2 par%beta  = 30.0    ! elevation angle [deg]
3 par%distance = 100.0

```

par%gamma sets the rotation of the detector plane about the line of sight. If par%distance is omitted it defaults to $100 \times r_{\max}$ (or the maximum box half-extent).

9.2 Multiple Observers

Multiple observers are specified as arrays:

```

1 par%alpha = 0.0  45.0  90.0  135.0
2 par%beta  = 0.0   0.0   0.0   0.0

```

Up to 99 simultaneous observers are supported.

9.3 Image Parameters

Parameter	Default	Description
par%nxim, par%nyim	0	Image pixel counts; peel-off enabled when > 0
par%dxim, par%dyim	auto	Pixel angular size [deg] (defaults to cover the full grid)

When par%dxim/par%dyim are not set, the pixel size is chosen to cover the full sphere:

$$\delta\theta = \frac{1}{N_x/2} \arcsin\left(\frac{r_{\max}}{d}\right). \quad (4)$$

9.4 Peel-off Output Flags

Parameter	Default	Description
par%save_peeloff	.false.	Save 1-D peel-off spectra per observer
par%save_peeloff_2D	.false.	Save peel-off spectral image cubes (x - y - ν)
par%save_peeloff_3D	.false.	Save full 3-D peel-off data
par%save_sightline_tau	.false.	Compute deterministic sight-line maps from each observer pixel: gas $\tau(\nu)$, N_{HI} , and dust τ (when par%DGR > 0). See §9.5.

9.5 Sight-line Optical-depth and Column-density Maps

When par%save_sightline_tau = .true., LaRT integrates the opacity along each observer pixel's line of sight and writes a separate FITS file <base>_tau<obs>.fits.gz for every observer. The file contains:

HDU	Content
TAU_gas	gas optical depth $\tau(\nu, x, y)$
N_gas	gas column density $N_{\text{HI}}(x, y)$
TAU_dust	dust optical depth $\tau_{\text{dust}}(x, y)$ (only when <code>par%DGR > 0</code>)

The pixel grid uses the same WCS as the peel-off image (`par%nxim`, `par%nyim`, `par%dxim`, `par%dyim`); the frequency axis matches the spectrum output (`par%nxfreq`, `par%velocity_min`, `par%velocity_max`). All three grid modes (Cartesian, AMR, clump) are supported with the appropriate raytrace dispatch.

Standalone driver `make_sightline_tau.x`

The same sight-line integration is exposed as a standalone executable that skips the photon-transport stage entirely:

```

1 make sightline_tau                                # builds make_sightline_tau.x
2 mpirun -np N ./make_sightline_tau.x sl.in          # writes <base>_tau.fits.gz

```

`sl.in` is an ordinary LaRT input file; the driver internally sets `par%save_peeloff` and the sight-line flag to `.true.` and otherwise honors every namelist parameter. Useful for parameter sweeps where only the deterministic maps are needed (no Monte-Carlo transport). Unlike `make_clumps.x`, multiple MPI ranks do speed this driver up: pixels are divided across `par%nproc` via `loop_divide` and reduced with `reduce_mem`.

The folder `examples/sightline_tau/` contains a reference set of six runs that exercise every grid mode and observer configuration the driver supports:

File	Grid mode	Observer
<code>car_sightline.in</code>	Cartesian	outside (rectangular FITS)
<code>amr_sightline.in</code>	AMR	outside (rectangular FITS)
<code>clump_sightline.in</code>	clumpy sphere	outside (rectangular FITS)
<code>car_inside.in</code>	Cartesian	inside (HEALPix all-sky)
<code>amr_inside.in</code>	AMR	inside (HEALPix all-sky)

The shipped `run.sh` executes all six in sequence; `make_amr_sphere_data.py` regenerates the AMR data file from a uniform-sphere physics model. The companion notebook `plot_sightline_tau.ipynb` displays the column-density maps, selected $\tau(x, y, \nu)$ slices, the all-sky HEALPix mollview maps and the sky-mean line profile.

tau_huge cap (sight-line vs. photon transport). The AMR raytrace routines used by the photon-transport stage early-exit when the running optical depth exceeds the double-precision $\exp(-\tau)$ underflow point ($\tau_{\text{huge}} \approx 745.2$); accumulating further contributes nothing to the photon's transmission. The sight-line variant `raytrace_to_edge_tau_gas_amr` exists precisely to *report* the integrated τ , however large, and therefore runs without that cap (matching the Cartesian `raytrace_to_edge_tau_gas` in `raytrace_car.f90`). The same applies to `raytrace_to_dist_tau_gas_amr` used by the inside-observer HEALPix path.

9.6 Inside Observer (HEALPix All-Sky)

Setting `par%inside > 0` switches the entire output and peel-off chain to the *inside-observer* path: the observer is placed at a chosen position (o_x, o_y, o_z) *inside* the medium, and each escaping photon is binned into an all-sky HEALPix map by the direction from the observer back to the photon. This mode is intended for “galactic” observations – e.g., placing the observer at the Sun’s position inside a model of the Galactic ISM, or inside a simulated galaxy.

Activation:

```

1 par%inside = 8                ! HEALPix Nside parameter; total npix = 12*inside^2
2 par%obsx   = 0.3              ! observer position (code units)
3 par%obsy   = 0.0
4 par%obsz   = 0.0
5 par%save_peeloff = .true.
6 par%save_peeloff_2D = .true. ! also save the 2D map (sum over frequency)

```

When `par%inside > 0`, `setup.f90` automatically sets `par%observer_located_inside = .true.` and switches:

- the observer arrays from rectangular images to HEALPix all-sky maps with $N_{\text{pix}} = 12N_{\text{side}}^2$;
- the peel-off routines to the `_inside` variants (which use `vec2pix` to bin the peeled weight by observer-pointing direction);
- `output_reduce`, `output_normalize` and `write_output` to the `_inside` variants.

In v2.10 the dispatch is performed by the type-bound methods of `grid_base_type` (`grid_class.f90`); the `grid_car_type` and `grid_amr_type` concrete types each branch on `par%observer_located_inside` and forward to the appropriate `_inside` or `_outside` concrete routine.

Multiple observers. As with the rectangular path, up to 99 observer positions are supported by passing arrays:

```

1 par%inside = 8
2 par%obsx   = 0.0  0.3  0.0
3 par%obsy   = 0.0  0.0  0.5
4 par%obsz   = 0.0  0.0  0.0

```

Output files. In *inside-observer* mode the `_obs.fits.gz` file holds 2D HEALPix arrays of shape `(npix, nxfreq)` with extensions `Direct` (unscattered direct contribution) and `Scattered` (scattered contribution). When `par%save_peeloff_2D = .true.` a separate `_obs2D.fits.gz` file is written with shape `(npix,)` holding the same arrays summed over frequency. The HEALPix maps use the RING ordering convention. The Spectrum HDU (`Jout`, `Jin`, `Jabs`) is always written regardless of observer mode.

Source scope. The *inside-observer* path supports the same internal sources as the standard path – `point`, `uniform`, `gaussian`, `exponential`, `seraic`, `diffuse_emissivity` – but *not* the external-illumination geometries (`point_illumination`, `stellar_illumination`, `plane_illumination`), which are only physically meaningful for an observer outside the medium. Stokes polarization is not yet implemented for the *inside-observer* path.

Grid-mode support. The HEALPix inside-observer path is available for both Cartesian (`par%grid_type = 'car'`) and AMR (`par%grid_type = 'amr'`) grids. In AMR mode, dedicated `_inside_amr` peeling, output and sight-line routines are used (see the *LaRT AMR Description* document for implementation details). Clumpy medium (`par%grid_type = 'clump'`) is not currently supported with the inside-observer path.

Sightline-tau output. When the inside-observer path is active, an additional `_sightline.fits.gz` file is produced that records τ and $N(\text{HI})$ from the observer position to each of the six box faces ($\pm x, \pm y, \pm z$). This is useful for cross-checking the optical depth structure seen by the observer.

Smoke test. The `examples/amr_sphere_generic/` directory ships `sphere_amr_inside_test.in` (AMR uniform sphere) and `sphere_car_inside_test.in` (Cartesian 64^3 counterpart), both with `par%nside = 8` and an off-center observer at $(o_x, o_y, o_z) = (0.3, 0, 0)$. These produce equivalent HEALPix maps within the documented AMR octree volume-averaging tolerance.

10. Output Files

Outputs are written either as gzip-compressed FITS files (`.fits.gz`) or as HDF5 files (`.h5`). HDF5 is the default; set `par%file_format = 'fits'` in the namelist to switch. Both formats share the same logical layout (one section per FITS HDU $\leftrightarrow$ one HDF5 group), so the conversion is loss-less and runs in both directions via `python/lart_io.py` (see §11. for the schema and converter, §15. for the Python helper).

10.1 Primary Output Arrays

Array	FITS extension	Description
Jout	JOUT	Emergent spectrum integrated over all escape directions
Jin	JIN	Injected (intrinsic) source spectrum
Jabs	JABS	Spectrum of photons absorbed by dust
Jmu	JMU	Emergent spectrum binned by $\mu = \cos \theta_z$

Jout is the main science output. It is normalized per unit frequency interval and per unit area of the bounding sphere. In dimensionless units (no `par%distance_unit`):

$$J_{\text{out}}(x) = \frac{\text{escaped photon weight in bin } x}{n_{\text{ph}} \Delta x 2\pi \pi R^2}, \quad (5)$$

where $R = r_{\text{max}}$ (Cartesian) or $R = L_{\text{box}}/2$ (AMR), and $\Delta x = 1$ in units of the Doppler width.

Jin requires `par%save_Jin = .true.`

Jabs requires `par%save_Jabs = .true.`

Jmu requires `par%save_Jmu = .true.`; see §10.2.

The total flux is conserved: $\int J_{\text{out}} dx + \int J_{\text{abs}} dx = \int J_{\text{in}} dx$ (in the absence of other losses).

10.2 Angular Spectrum Jmu

Setting `par%save_Jmu = .true.` adds a 2-D image extension `EXTNAME='Jmu'` of shape (n_x, n_μ) that records the emergent spectrum binned by the z -component of the escape direction, $\mu \equiv \cos \theta_z = \hat{k}_z$. This is useful for

- verifying isotropy of homogeneous models (every μ bin should match `Jout` within Poisson noise),
- characterizing the angular dependence of escape in non-isotropic models (for example, a slab or rotating halo).

Binning.

- `par%nmu` bins of width $\Delta\mu$ tile the μ range uniformly.
- Default `par%nmu = 11` (odd) places $\mu = 0$ at a bin center, so the perpendicular-escape spectrum is preserved as a single channel. Even `par%nmu` places $\mu = 0$ at a bin edge and cleanly separates the $+z$ and $-z$ hemispheres.
- Range:
 - `par%xyz_symmetry = .true.` folds escapes onto the $+z$ hemisphere, so $\mu \in [0, +1]$ and $\Delta\mu = 1/n_\mu$.
 - Otherwise $\mu \in [-1, +1]$ and $\Delta\mu = 2/n_\mu$.

The header keywords `nmu`, `mu_min`, `dmu` record these values, and full WCS keywords (`CTYPE1='XFREQ'`, `CTYPE2='MU'`, `CRVAL*`, `CDELTA*`) are written to the image header.

Normalization. `Jmu` uses the same denominator as `Jout` multiplied by n_μ , so that

$$\frac{1}{n_\mu} \sum_{i=1}^{n_\mu} J(x; \mu_i) = J_{\text{out}}(x) \quad (6)$$

holds exactly. Equivalently, in a homogeneous, isotropic medium each μ bin reproduces `Jout` up to Poisson fluctuations, providing a cheap regression test for new geometries.

Modes. Available with all three grid modes: Cartesian, AMR octree, and the clumpy medium.

10.3 Peel-off cube units and conversion to Jmu units

When peel-off is enabled (`par%nxim`, `par%nyim` > 0 and `par%save_peeloff_3D = .true.`), each observer's `<base>_obs_NNN.fits.gz` file contains Scattered and Direct 3-D image cubes, plus optionally `Direct0`. These are written by `output_sum_rect.f90` as a *specific intensity per pixel*, normalized so that

$$I_{\text{peel}}(x, y, \nu) = \frac{\text{photon weight in pixel } (x, y) \text{ and bin } \nu}{n_{\text{ph}} \delta\Omega_{\text{pix}} \Delta\nu d_{\text{cm}}^2}, \quad (7)$$

where $\delta\Omega_{\text{pix}} = (\text{CD2_2} \cdot \text{CD3_3}) (\pi/180)^2$ is the per-pixel solid angle, $\Delta\nu$ is `Dxfreq` or `Dwave` (set by `par%intensity_unit`), d_{cm}^2 is the square of the distance unit in cm, and n_{ph} is the photon count.

Conversion to Jmu units. J_{out} and J_{mu} are normalized by $4\pi R^2$ (the source surface area) and the outward solid angle 2π , whereas the peel-off cube is normalized by the per-pixel solid angle and d_{cm}^2 . Summing the cube over pixels and rescaling brings the two onto a common axis:

$$J_{\text{peel}}(\nu) = \left(\sum_{(x,y)} I_{\text{peel}}(x,y,\nu) \right) \cdot \frac{4\pi D^2 \delta\Omega_{\text{pix}}}{A_{\text{surface}} \cdot 2\pi}, \quad (8)$$

where D is DISTANCE (in code units) and A_{surface} is the source surface area in code units, with the geometry-dependent forms mirroring `output_sum_rect.f90`:

Geometry	A_{surface} (code units)	Notes
sphere	$4\pi r_{\text{max}}^2$	default; clumpy medium also uses this
box (rectangle)	$8(x_{\text{max}} y_{\text{max}} + y_{\text{max}} z_{\text{max}} + z_{\text{max}} x_{\text{max}})$	full surface of $[-x_{\text{max}}, x_{\text{max}}] \times \dots$ box
slab (par%xy_periodic)	$2(2x_{\text{max}})(2y_{\text{max}})$	two unbounded boundaries
AMR octree	4π	equivalent to $r_{\text{max}} = 1$ (cf. §7.)

The bin width $\Delta\nu$ cancels because both J_{mu} and I_{peel} are already normalized per bin. At convergence and for an axisymmetric source,

$$J_{\text{peel}}(\nu; \alpha, \beta) \rightarrow J_{\text{mu}}(\nu; \mu = \cos \beta). \quad (9)$$

The Python helper `LaRTOutput.plot_peel_jmu_compare()` (in `python/read_lart.py`) implements Eq. (8) with the surface area chosen automatically from `par%geometry`, `par%rmax`, and the spectrum-header `XMAX/YMAX/ZMAX/XY_PER`. See also `examples/sphere/plot_Jpeel_conversion.ipynb` and `examples/clump_sphere/verify_peel_jmu.ipynb`.

Component decomposition (clumpy medium). For a point source inside a clumpy sphere with covering factor f_{cov} , the fraction of photons that escapes without scattering is $\exp(-f_{\text{cov}})$ (Poisson statistics), and the fraction that scatters at least once is $1 - \exp(-f_{\text{cov}})$. J_{mu} tallies *both* populations. The peel-off cube splits them:

- Scattered. The MC estimator over many scatter events averages efficiently, so

$$\int J_{\text{peel}}^{\text{scatt}} d\nu \approx (1 - e^{-f_{\text{cov}}}) \int J_{\text{mu}} d\nu, \quad (10)$$

and the equality holds across all observer angles within Monte-Carlo noise.

- Direct. For a *point* source, every photon shares the same line of sight to a given observer, so the direct contribution is the binary realization of $\exp(-\tau_{\text{LOS}})$: either ~ 1 (clear LOS) or ~ 0 (blocked by an intervening clump) per observer. J_{mu} , by contrast, is azimuth averaged and captures direct escapes uniformly. Adding the binary Direct and the smooth Scattered therefore only matches J_{mu} *in expectation* – it can deviate substantially per observer at finite photon counts.

For shape comparisons in clumpy runs, prefer the scattered component (`component='scatt'` in `plot_peel_jmu_compare()`) so that Eq. (10) provides a clean, deterministic relation.

10.4 Output Control Parameters

Parameter	Default	Description
par%file_format	'hdf5'	Output container: 'hdf5' ($\rightarrow$.h5) or 'fits' ($\rightarrow$.fits.gz); see §11.
par%out_file	''	Output file name (defaults to <input>.h5; extension is corrected to match par%file_format with a warning)
par%out_merge	.false.	Accumulate into existing output instead of overwriting
par%save_backup	.false.	On merge, copy the prior output to <base>_NN.{h5,fits.gz} before overwriting
par%out_bitpix	0	Storage precision: 0=auto (dynamic-range driven), -32=float32, -64=float64
par%save_Jin	.false.	Save injected spectrum
par%save_Jabs	.false.	Save dust-absorbed spectrum
par%save_Jmu	.false.	Save angular spectrum $J(x; \mu)$ (§10.2)
par%nmu	11	Number of μ bins for Jmu
par%save_all	.false.	Save full 3-D $J(\nu, x, y, z)$ array
par%save_direct0	.false.	Save photon escape direction distributions
par%save_radial_profile	.false.	Save 1-D radial J profile
par%save_sightline_tau	.false.	Save optical depth along sightlines
par%intensity_unit	''	Physical unit label written to the FITS header

par%out_merge = .true. is useful for noise reduction: run the same setup multiple times and merge the results. All input parameters must be identical across runs.

11. File Formats: FITS and HDF5

LaRT writes one logical layout that lives in either FITS or HDF5 files. The choice is made by par%file_format (default 'hdf5'); the filename extension is corrected to match if the user-supplied par%out_file disagrees.

11.1 Format selection

- par%file_format = 'hdf5' (default) writes .h5 files via the hdf5io_mod backend. Build with HDF5=1 (the Makefile default).
- par%file_format = 'fits' writes .fits.gz files via the legacy fitsio_mod backend (CFITSIO). Works regardless of the HDF5 build option.

The mismatch warning is issued once on rank 0, e.g. if par%out_file = 'foo.fits.gz' is given alongside par%file_format = 'hdf5':

```
WARNING: par%out_file (foo.fits.gz) does not match
        par%file_format = hdf5
        Overriding par%out_file to: foo.h5
        (set par%file_format explicitly to suppress this warning.)
```

All derived filenames (peel-off _obs, sight-line _tau, _allph, backup) follow par%file_format as well, so the main output and its derived files always share a format.

11.2 HDF5 schema

Each FITS HDU maps to one HDF5 group; the HDU's header keywords become attributes on that group; the HDU's data is either a single /<name>/data dataset (image-style) or one dataset per BinTable column (table-style). Example for a sphere run with peel-off:

```
run.h5
/Spectrum                                     (group, table-style)
/Spectrum/Xfreq                             (1-D dataset, float32)
/Spectrum/velocity                          (1-D dataset, float32)
/Spectrum/wavelength                        (1-D dataset, float64)
/Spectrum/Jout                              (1-D dataset, float32)
/Spectrum/Jin                              (1-D dataset, float32)
/Spectrum/@nphotons                         (attribute)
/Spectrum/@taumax                           (attribute)
/Spectrum/@Nsc_gas                          (attribute)
...

run_obs.h5
/Scattered                                  (group, image-style)
/Scattered/data                            (3-D dataset, chunked + gzip)
/Scattered/@CD1_1                          (WCS attribute)
...

/Direct
/Direct/data
/Direct/@EXTNAME                           = 'Direct'

run_tau.h5
/TAU_gas/data, /N_gas/data, [/TAU_dust/data when DGR > 0]
```

Sections appear in their write order on both the root group and inside each subgroup (creation-order tracking is enabled), so h5py iteration returns columns and HDUs in the same order LaRT wrote them. Datasets larger than 4096 elements are chunked and compressed with gzip(4); smaller ones are stored contiguous. The on-disk storage type follows par%out_bitpix (auto / float32 / float64) just as for FITS.

11.3 Conversion utility

The Python helper `python/lart_io.py` reads either format and can convert between them losslessly:

```
python lart_io.py info run.h5
python lart_io.py convert run.fits.gz run.h5
python lart_io.py convert run.h5 run.fits.gz
```

The round-trip preserves all data bit-identically; only string keyword names are uppercased on the FITS leg, matching the FITS card-name convention.

11.4 HDF5 library requirements

LaRT links against the HDF5 Fortran library (`libhdf5_fortran.a` plus the C core `libhdf5.a`, version ≥ 1.14). The Fortran `.mod` files are compiler-specific: an HDF5 build linked against `gfortran` cannot be used with Intel `ifx` or `ifort`, and vice versa. Set `par%HDF5_PREFIX` on the make command line to choose a build that matches the LaRT compiler.

12. Dust and Polarization

Parameter	Default	Description
<code>par%DGR</code>	0.0	Dust-to-gas ratio relative to the Milky Way value
<code>par%albedo</code>	0.32536	Single-scattering albedo ω
<code>par%hgg</code>	0.67617	Heney–Greenstein asymmetry parameter g
<code>par%use_stokes</code>	<code>.false.</code>	Enable full Stokes polarization RT
<code>par%scatt_mat_file</code>	<code>''</code>	Mueller matrix file for polarized dust scattering
<code>par%use_reduced_wgt</code>	<code>.false.</code>	Reduce photon weight by albedo instead of destroying absorbed photons

When `par%DGR = 0` (default), there is no dust and all photon weight is conserved. Non-zero `par%DGR` adds dust extinction and Heney–Greenstein scattering. When `par%use_stokes = .true.`, a Mueller matrix file must be provided; files for the Weingartner–Draine Milky Way model are included in the `data/` directory.

13. Parallelism

Parameter	Default	Description
<code>par%no_photons</code>	10^6	Total photon packets (floating-point, converted to int64)
<code>par%use_master_slave</code>	<code>.true.</code>	Dynamic load balancing via master–worker algorithm
<code>par%num_send_at_once</code>	100	Photons dispatched to each worker per batch
<code>par%nprint</code>	10^5	Progress-report interval [photons]
<code>par%iseed</code>	auto	Random seed; defaults derived from MPI rank and system clock

With `par%use_master_slave = .true.`, one MPI rank acts as dispatcher and the rest as workers. This balances workloads when scattering rates vary across photon packets. With `.false.`, each rank processes an equal share of `par%no_photons`.

14. Complete Parameter Reference

Table 1 lists commonly used parameters. Parameters marked [†] are new or changed in v2.10.

Table 1: LaRT parameter reference.

Parameter	Default	Description
<i>Simulation control</i>		
<code>par%no_photons</code>	10^6	Number of photon packets
<code>par%nprint</code>	10^5	Progress interval [photons]
<code>par%iseed</code>	auto	Random seed
<i>Cartesian grid</i>		
<code>par%nx, par%ny, par%nz</code>	101	Cells per axis
<code>par%xmax, par%ymax, par%zmax</code>	1.0	Half-box size (code units)
<code>par%rmax</code>	-999	Spherical boundary radius
<code>par%rmin</code>	-999	Inner cavity radius
<code>par%geometry</code>	'sphere'	J output geometry (§4.2)
<code>par%xyz_symmetry</code>	.false.	+x, +y, +z octant only
<code>par%xy_symmetry</code>	.false.	+x, +y quadrant only
<code>par%xy_periodic</code>	.false.	Periodic in x, y
<code>par%distance_unit</code>	' '	Physical unit ('kpc', 'pc', 'au')
<code>par%distance2cm</code>	1.0	1 code unit in cm
<i>Grid mode</i>		
<code>par%grid_type[†]</code>	'car'	v2.10+: 'car', 'amr', 'clump'
<code>par%use_amr_grid</code>	.false.	v2.00 AMR flag
<code>par%use_clump_medium</code>	.false.	v2.00 clump flag
<i>Source</i>		
<code>par%source_geometry</code>	'point'	See §5.1
<code>par%xs_point, par%ys_point,</code> <code>par%zs_point</code>	0.0	Point source position
<code>par%source_rmax</code>	-999	Source truncation radius
<code>par%source_rscale</code>	-999	Radial scale length

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(continued)

Parameter	Default	Description
par%source_zscale	0.0	Vertical scale height
par%sersic_m	4.0	Sérsic index
par%Reff	1.0	Effective radius
par%comoving_source	. true.	Source co-moving with local gas
par%emiss_file	''	Emissivity FITS cube
par%star_file	''	Point-star list (x, y, z, L)
<i>Line and spectrum</i>		
par%line_id	$\text{Ly}\alpha$	Atomic transition (§5.3)
par%spectral_type	'monochromatic'	Emission frequency distribution
par%temperature	10^4 K	Gas temperature
par%temperature0	$= T$	Source temperature for voigt0
par%bturb	0.0	Turbulent dispersion [km s^{-1}]
par%gaussian_width_vel	0.0	Gaussian line width [km s^{-1}]
par%xfreq0	0.0	Frequency offset for emitted photons
par%line_prof_file	''	Custom line profile file
par%line_prof_file_type	0	0: (x, I); 1: (λ, I)
<i>Frequency grid</i>		
par%nxfreq	121	Spectral bins
par%xfreq_min, par%xfreq_max	auto	x -grid limits
par%nwavelength	0	Wavelength bins (overrides x -grid)
par%wavelength_min,	—	[Å]
par%wavelength_max		
par%nvelocity	0	Velocity bins (overrides x -grid)
par%velocity_min,	—	[km s^{-1}]
par%velocity_max		
<i>Optical depth and density</i>		
par%taumax	—	Line-center τ_0 : center $\rightarrow +z$ boundary
par%tauhomo	—	Same, uniform-density equivalent
par%N_HI_max	—	Column density: center $\rightarrow +z$ [cm^{-2}]
par%N_HI_homo	—	Same, uniform-density equivalent

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(continued)

Parameter	Default	Description
par%density_rscale	-999	Exponential density scale radius
<i>External data</i>		
par%dens_file	''	Density FITS cube [cm^{-3}]
par%temp_file	''	Temperature FITS cube [K]
par%velo_file	''	Velocity FITS cube [km s^{-1}]
par%input_field	''	Base name for .dens/.temp/.velo.fits.gz
<i>Velocity</i>		
par%velocity_type	''	Velocity law (§6.4)
par%Vexp	0.0	Expansion/infall speed [km s^{-1}]
par%Vx, par%Vy, par%Vz	0.0	Uniform velocity [km s^{-1}]
par%Vpeak	0.0	Peak velocity for SSH [km s^{-1}]
par%rpeak	0.0	Radius at peak for SSH
par%DeltaV	0.0	Velocity increment for SSH [km s^{-1}]
par%Vrot	0.0	Rotation speed [km s^{-1}]
par%recoil	.false.	Photon recoil in scattering
<i>AMR</i>		
par%amr_type	'ramses'	'generic' or 'ramses'
par%amr_file	''	AMR data path
par%amr_snapnum	1	RAMSES snapshot number
<i>Clumpy medium</i>		
par%clump_radius	—	Clump radius
par%clump_tau0	—	Line-center τ_0 per clump
par%clump_f_cov	—	Covering factor C_f
par%clump_f_vol	—	Volume filling factor f_v
par%clump_N_clumps	—	Number of clumps N
par%clump_sigma_v	0.0	Random velocity dispersion [km s^{-1}]
par%save_clump_info	.false.	Write clump table to FITS
<i>Observers and peel-off</i>		

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(continued)

Parameter	Default	Description
par%nxim, par%nyim	0	Image pixels; peel-off active when > 0
par%dxim, par%dyim	auto	Pixel size [deg]
par%distance	$100 r_{\max}$	Observer–center distance
par%obsx, par%obsy, par%obsz	(0, 0, 1)	Observer direction
par%alpha, par%beta	—	Azimuth, elevation [deg]
par%gamma	0.0	Detector-plane rotation [deg]
par%save_peeloff	.false.	1-D peel-off spectra
par%save_peeloff_2D	.false.	Peel-off spectral images
par%save_peeloff_3D	.false.	Full 3-D peel-off cube
par%save_direc0	.false.	Peel-off image without $e^{-\tau}$ (transparent-medium reference)
<i>Output</i>		
par%file_format	'hdf5'	Output container: 'hdf5' or 'fits'
par%out_file	''	Output file name (extension follows par%file_format)
par%out_merge	.false.	Accumulate into existing file
par%save_backup	.false.	Keep _NN copy of prior output on merge
par%out_bitpix	0	Storage precision: 0 = auto, -32, -64
par%save_Jin	.false.	Save injected spectrum
par%save_Jabs	.false.	Save absorbed spectrum
par%save_Jmu	.false.	Save angular spectrum $J(x; \mu)$
par%nmu	11	Number of μ bins for Jmu
par%save_all	.false.	Save full 3-D $J(\nu, x, y, z)$
par%save_radial_profile	.false.	Save 1-D radial J
par%save_sightline_tau	.false.	Save sightline τ
par%intensity_unit	''	Output header unit label
<i>Dust</i>		
par%DGR	0.0	Dust-to-gas ratio
par%albedo	0.32536	Single-scattering albedo

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(continued)

Parameter	Default	Description
par%hgg	0.67617	Henyeey–Greenstein asymmetry g
par%use_stokes	.false.	Polarization RT
par%scatt_mat_file	''	Mueller matrix file
par%use_reduced_wgt	.false.	Reduce weight instead of destroy
<i>Parallelism</i>		
par%use_master_slave	.true.	Dynamic load balancing
par%num_send_at_once	100	Photons per worker batch

15. Python Utilities

A small set of post-processing scripts is shipped under `python/` in each version's source tree. They depend on `numpy`, `astropy`, and (for plotting scripts) `matplotlib`.

15.1 Reading FITS output: `read_lart.py`

`read_lart.py` is a thin reader that loads the spectral arrays from a LaRT FITS output. It accepts the input-file name in any of three forms:

- full input file: `t1tau2.in`
- stem only: `t1tau2` (`.in` is added automatically)
- the FITS file: `t1tau2.fits.gz`

The script honors `par%out_file` when set in the namelist; otherwise it falls back to `<stem>.fits.gz` in the same directory as the input file.

Module API.

```

1 from read_lart import read_lart
2 out = read_lart('t1tau2.in')
3
4 # 1-D arrays, length nxfreq
5 out.xfreq, out.velocity, out.wavelength
6 out.Jout          # always present
7 out.Jin, out.Jabs, out.Jabs2    # optional -- None if not in file
8
9 # Jmu output -- present only if par%save_Jmu = .true. was used
10 out.mu            # bin centers, length nmu
11 out.mu_edges      # bin edges, length nmu+1

```

```

12 out.Jmu          # shape (nmu, nxfreq); None if absent
13 out.nmu, out.mu_min, out.dmu
14
15 # Raw FITS headers
16 out.spectrum_header, out.jmu_header

```

Every optional array is None when its FITS column or HDU is not present, so the reader works on outputs from any LaRT version.

Plotting Jmu. `LaRTOutput.plot_jmu()` draws the angular spectrum directly:

```

1 out = read_lart('t1tau3.in')
2 out.plot_jmu(kind='lines', x='velocity') # one curve per mu bin
3 out.plot_jmu(kind='image', x='velocity') # 2-D heatmap (x vs mu)

```

The `x=` keyword selects the abscissa ('velocity', 'xfreq', or 'wavelength'); the `kind=` keyword toggles between line ensemble (colored by μ) and a 2-D image. The line view also overplots `Jout` as a black dashed reference. Pass `ax=` to draw onto an existing axes, `savefig='out.png'` to save to disk. Axis limits accept either the tuple form `xlim=(lo,hi)` / `ylim=(lo,hi)` or the per-bound shorthand `xmin/xmax/ymin/ymax`; the per-bound override takes precedence when both are given:

```

1 out.plot_jmu(kind='lines', xlim=(-500, 500), ymin=0)
2 out.plot_jmu(kind='lines', xmax=300) # left edge auto-scaled

```

Command-line usage.

```

1 python read_lart.py t1tau2          # auto-resolves to t1tau2.in
2 python read_lart.py t1tau2.in
3 python read_lart.py t1tau2.fits.gz

```

prints a one-screen summary of which arrays are present and, when `Jmu` is in the file, the per-bin $\max_x J(x; \mu) / \max_x J_{\text{out}}(x)$ ratio (a quick isotropy sanity check for homogeneous models).

Other scripts. The following plotting helpers are also shipped:

- `plot_spectrum.py`, `plot_map.py`, `plot_intensity_profile.py`, `plot_Pa.py`: standard quick-look plots.
- `check_flux.py`, `flux_check.py`: flux-conservation diagnostics.
- `AMR_grid/`: helpers for generating generic AMR data files (see `AMR_grid/make_amr_grid.py`, `make_amr_sphere_radial.py`).

15.2 Format-agnostic reader and FITS ↔ HDF5 converter: `lart_io.py`

`python/lart_io.py` provides a single `load_lart(path)` entry point that reads either a FITS or an HDF5 LaRT output and returns a uniform `LartFile` structure (one Section per HDU / group, with `.data` or `.columns` plus `.attrs`). It

also converts between the two formats by extension. Requires numpy plus astropy (for FITS) and/or h5py (for HDF5); each backend is only imported when needed.

Module API.

```

1 from lart_io import load_lart, convert
2
3 f = load_lart('run.h5')           # works on .fits.gz too
4 spec = f['Spectrum']             # a Section
5 jout = spec.columns['Jout']       # 1-D numpy array
6 npht = spec.attrs['nphotons']
7
8 scat = load_lart('run_obs.h5')['Scattered']
9 cube = scat.data                 # 3-D array (nxim, nyim, nxfreq)
10
11 convert('run.fits.gz', 'run.h5') # both directions by extension
12 convert('run.h5', 'run.fits.gz')
```

The round-trip preserves all numerical data bit-identically; string keyword names are normalized to uppercase on the FITS leg (FITS card-name convention), which is the only visible difference between the two forms.

Command-line usage.

```

1 python lart_io.py info run.h5
2 python lart_io.py info run.fits.gz
3 python lart_io.py convert run.fits.gz run.h5
4 python lart_io.py convert run.h5 run.fits.gz
```

16. Example Input Files

16.1 Uniform Sphere, $\text{Ly}\alpha$, Static

```

1 &parameters
2   par%no_photons      = 1e6
3   par%temperature     = 1.0e4
4   par%taumax         = 1.0e4
5   par%rmax           = 1.0
6   par%nx = 101, par%ny = 101, par%nz = 101
7   par%source_geometry = 'point'
8   par%spectral_type   = 'voigt'
9   par%nxfreq          = 121
10  /
```


16.2 Sphere with Hubble Outflow and Peel-off

```

1 &parameters
2   par%no_photons      = 1e6
3   par%temperature     = 1.0e4
4   par%taumax          = 1.0e4
5   par%rmax            = 1.0
6   par%nx = 101, par%ny = 101, par%nz = 101
7   par%source_geometry = 'point'
8   par%spectral_type    = 'voigt'
9   par%velocity_type    = 'hubble'
10  par%Vexp             = 200.0          ! km/s outflow
11  par%nxfreq           = 121
12  par%nxim = 129, par%nyim = 129
13  par%alpha = 0.0  45.0  90.0
14  par%beta  = 0.0   0.0   0.0
15  par%save_peeloff_2D = .true.
16 /

```

16.3 Realistic Galaxy (FITS Density Field, Stokes Polarization)

```

1 &parameters
2   par%no_photons      = 1e8
3   par%dens_file       = 'galaxy.dens.fits.gz'
4   par%temp_file       = 'galaxy.temp.fits.gz'
5   par%velo_file       = 'galaxy.velo.fits.gz'
6   par%taumax          = 1.0e4          ! rescale to this optical depth
7   par%distance_unit   = 'kpc'
8   par%source_geometry = 'exponential'
9   par%source_rscale   = 3.0
10  par%source_zscale   = 0.5
11  par%spectral_type    = 'voigt'
12  par%DGR              = 1.0
13  par%use_stokes       = .true.
14  par%nxfreq           = 121
15  par%nxim = 200, par%nyim = 200
16  par%save_peeloff_2D = .true.
17 /

```

16.4 AMR Sphere (Generic Text Format)

```

1 &parameters
2   par%use_amr_grid    = .true.           ! or par%grid_type = 'amr' in v2.10
3   par%amr_type        = 'generic'
4   par%amr_file         = 'sphere_amr.dat'
5   par%no_photons       = 1e6
6   par%taumax           = 1.0e4
7   par%source_geometry = 'point'
8   par%spectral_type    = 'voigt'
9   par%nxfreq           = 121
10  par%nxim = 100, par%nyim = 100
11  par%save_peeloff_2D = .true.
12  par%save_Jin          = .true.
13 /

```

16.5 Clumpy Sphere with Random Velocities

```

1 &parameters
2   par%use_clump_medium = .true.           ! or par%grid_type = 'clump' in v2.10
3   par%rmax              = 1.0
4   par%clump_radius       = 0.01
5   par%clump_f_cov        = 5.0
6   par%clump_tau0         = 1.0e3
7   par%temperature        = 1.0e4
8   par%clump_sigma_v      = 50.0           ! km/s random dispersion
9   par%no_photons         = 1e6
10  par%spectral_type       = 'voigt'
11  par%nxfreq              = 301
12  par%velocity_min        = -200.0
13  par%velocity_max        = 200.0
14  par%save_clump_info     = .true.
15 /

```